

PRIME DNA

The Odd Weil Form on a Window

The sliding-window identity, the logarithmic edge law,
and the bounded-relay conjecture

Revised after independent review

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Abstract

Let $\lambda(a)$ be the floor of the odd-sector Weil form over test functions supported in $[-a, a]$. Weil's criterion, in Yoshida's odd-test form, says that $\lambda(a) \geq 0$ for every a is equivalent to the Riemann Hypothesis; the floor is certified positive to $a = 0.8$ (Zhu) and numerically decays doubly exponentially in a . We study the floor as a function of the window. Two results are exact. First, the archimedean part of the form on a window is one half of the Dirichlet form of the logarithmic Laplacian plus a bounded form, with an explicit confining wall $\frac{1}{2} \log(1/(a - |y|))$ at the edge; by the boundary theory of the logarithmic Laplacian, a bounded minimizer is bounded by $C_0(\log(a/\delta))^{-1/2}$ at distance δ from the edge, and numerically it behaves like $C(a)(\log(a/\delta) + \beta)^{-1/2}$ with an amplitude $C(a)$ that we measure on meshes graded to 10^{-12} of the edge. Consequently the entry of a prime power into the window produces no kink in λ but a soft kink, λ' continuous with an increment of order $C^2/\log(1/\varepsilon)$ at scale ε , which we verify over four decades. Second, between entries the floor obeys the sliding-window identity

$$a \lambda'(a) = - \left[D_\infty(f) + D_P(f) + 2P(f)^2 + 2P(f)M(f) \right],$$

proved for the K -mode form and, for the exact form, as a two-sided Dini-derivative statement holding at every point of differentiability, where f is the minimizer, D_∞ is the archimedean dilation form, whose symbol tends to 1, $D_P(f) = - \sum_n 2\Lambda(n)n^{-1/2} \log n g'_f(\log n)$ is the Euler product differentiated against the derivative of the autocorrelation of f at the prime-power positions, and P, M are the polar moments. We evaluate its three terms in ball arithmetic on the K -mode minimizer and confirm the identity independently on the Fourier side; three order-one quantities cancel to the size of λ . The floor is continuous and non-increasing, and under a bounded-variation hypothesis on near-minimizers it is locally Lipschitz; every minimizer computed, on meshes graded to 10^{-12} of the edge and with up to 160 modes, has total variation exactly $4 \sup |f|$, being unimodal on each half window, so the regularity clause of the conjecture below reduces to that hypothesis. A third statement is numerical: on seven supports, with and without primes, $\lambda'(a) = -2C(a)^2$ to a part in a thousand (an eighth, at an entry, to 1.7%), the Hadamard formula one expects for this window problem, so that $\Phi' = 2C^2/\lambda$ with $\Phi = -\log \lambda$. The derivative of the floor is a property of the ground state alone: the pencil $(-Q', Q)$ has eigenvalues of size $6 \times 10^{10} T^*$ at $a = 1$ while the minimizer's direction gives $3.7 T^*$, so no inequality between the two forms can hold. With the horizon $T^*(a) = 2\pi e^{2a}$, finite-mode upper bounds on the floor from height 17 to height 130 (down to $\lambda \approx 10^{-93}$, with up to 230 modes and

rigorous enclosures of the truncated matrix's lowest eigenvalue beyond $a = 1$) give Φ'/T^* rising from 3.2 to 3.9 on coarse grids, and the Hellmann–Feynman derivative resolves a sawtooth at every prime-power entry: a rise into the entry, of 10 to 45% over the last 0.003, as the form without the entering prime heads toward indefiniteness, a drop across it, and a relaxation after, with peaks 3.3 at the entry of 2 and 4.1–4.3 at the entries of 3, 4, 5, 7. We state the bounded-relay conjecture, λ locally absolutely continuous with $\lambda' \geq -cT^*\lambda$ almost everywhere, which implies the Riemann Hypothesis by Grönwall's inequality from Zhu's certified support, and the decay law $\Phi'/T^* \rightarrow c_0$; the data exclude the raw phase-space count, disfavour the logarithmic law and the derivative of Zhu's even-sector Landau–Widom law, and favour a constant near 3.8. Evaluated on the first 6000 zeros of ζ , the sum of $|F(\gamma)|^2$ over the zeros returns the floor of the K -mode minimizer to 3×10^{-4} or better at $a = 0.6, 0.8, 1.0$, down to $\lambda = 10^{-26}$, and the zero-side dilation sum returns its derivative to 3×10^{-3} or better: the explicit formula closes on the actual zeros for both, a third and independent measurement of the identity after the matrices and the Fourier symbol. The zeros below the horizon carry at most 10^{-5} of the floor, the first zero 10^{-24} of it at $a = 1$, and the floor is made by the zeros between one and a half and three horizons. Under the Riemann Hypothesis the floor is a sampling constant for the zeros above the horizon, and a prolate-type count bounds Φ below by about half of what is observed. Nothing in this paper proves the Riemann Hypothesis. Theorems, computations with their precision, and conjectures are distinguished throughout.

Standalone research paper. Companion to Primes, Folds, and the One Dot and to Theta Kernels, Weil Positivity, and the Mathematical Theory Atlas. Revised after an independent referee report; the report's corrections are incorporated and acknowledged in Section 9.

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Status conventions

A **Theorem**, **Proposition** or **Lemma** without citation is proved in the text. A statement with a citation is taken from the literature and its hypotheses are stated. A **Computation** is a numerical result with its method, precision and cross-checks; a finite-mode minimum is a minimum over a

subspace and hence an upper bound on the floor, and every such value is labelled by its mode count K ; derivatives of finite-mode floors are Hellmann–Feynman derivatives of the K -mode form and are called so, never “exact”. A **Conjecture** is a precise statement with the evidence for it. A **Heuristic** or **Remark** makes no mathematical claim. The hypothesis that a minimizer is bounded, needed for the edge law, is written (H_∞) ; the hypothesis that near-minimizers have bounded sup norm times total variation, which gives Lipschitz continuity of the floor, is written (H_{BV}) ; both are carried explicitly wherever they are used.

1 The form, the floor and the horizon

For $f \in L^2(\mathbb{R})$ real with compact support write $F = \hat{f}$, $F(t) = \int f(x)e^{itx} dx$, and $g = f \star \tilde{f}$, $g(u) = \int f(x)f(x-u) dx$, so that $\hat{g} = |F|^2$. The Weil form in the normalization used throughout is

$$Q(f) = W_\infty(g) - \sum_{n \geq 2} \Lambda(n)n^{-1/2}(g(\log n) + g(-\log n)) + (\hat{g}(i/2) + \hat{g}(-i/2)), \quad (1)$$

$$W_\infty(g) = -(\gamma + \log \pi)g(0) + 2 \int_0^\infty \frac{g(0)e^{-2x} - g(x)e^{-x/2}}{1 - e^{-2x}} dx. \quad (2)$$

By the explicit formula [1], $Q(f) = \sum_\rho \hat{g}((\rho - \frac{1}{2})/i)$ over the nontrivial zeros; on the geometric side, in Fourier form,

$$Q(f) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \Psi(t) dt + \text{polar}, \quad \Psi(t) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi - \sum_n \frac{2\Lambda(n)}{\sqrt{n}} \cos(t \log n), \quad (3)$$

where the polar term is $2F(i/2)^2$ for even f and $-2P(f)^2$, $P(f) = \int f(x) \sinh(x/2) dx$, for odd f ; the multiplier $\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$ is that of W_∞ by the integral representation of ψ . This is the normalization of Zhu [5] and of Checkpoints 15–20 of [17]; Computation 2.1 confirms the identification numerically.

Definition 1.1. For $a > 0$ the *odd floor* is

$$\lambda(a) = \inf\{Q(f) : f \text{ odd, } \operatorname{supp} f \subset [-a, a], \|f\|_2 = 1, Q(f) < \infty\},$$

$\Phi(a) = -\log \lambda(a)$ wherever $\lambda(a) > 0$, and the *horizon* is $T^*(a) = 2\pi e^{2a}$. $N(T)$ denotes the number of zeros with $0 < \gamma \leq T$.

Since g is supported in $[-2a, 2a]$, only the prime powers with $\log n < 2a$ enter $Q(f)$; the *entries* are $a_n = \frac{1}{2} \log n$. The form Q itself does not depend on a , so λ is non-increasing.

Theorem 1.2 (Weil [1]; Yoshida [2], Prop. 1, as recorded in Suzuki [4]). *The Riemann Hypothesis holds if and only if $Q(f) \geq 0$ for every compactly supported smooth f ; and $Q(f) \geq 0$ for every odd such f already implies it. Hence $\text{RH} \iff \lambda(a) \geq 0$ for all $a > 0$.*

Theorem 1.3 (Zhu [5], Theorem 6.2). *For $a = 0.8$, in the odd sector, $8.206 \times 10^{-15} \leq \lambda(0.8) \leq 2.347 \times 10^{-14}$, certified by interval arithmetic (and in the even sector $8.9 \times 10^{-18} \leq \lambda^{\text{even}}(0.8) \leq 2.523 \times 10^{-16}$). In particular $\lambda(a) > 0$ for $0 < a \leq 0.8$.*

The horizon is the height at which the zeros become too dense for the window. A function of exponential type a has zeros of upper density at most a/π (Jensen’s formula; see Boas [11]); the zeros of ζ have density $(2\pi)^{-1} \log(t/2\pi)$ at height t , which equals a/π exactly at $t = T^*(a)$. Zhu’s resolution threshold, $2\pi e^{(4+o(1))e^a}$ for a one-stroke positivity certificate at half-width a [5], is a different and far larger scale. Everything below is organized around the pair $(\lambda(a), T^*(a))$.

Lemma 1.4 (existence, monotonicity, continuity, differentiability a.e.). *For every $a > 0$ the infimum defining $\lambda(a)$ is attained, the infimum over odd $C_c^\infty(-a, a)$ functions is the same number, λ is non-increasing and continuous, and λ is differentiable at almost every a .*

Proof sketch. By Proposition 3.1 and the discussion after it, Q on $L_{\text{odd}}^2(-a, a)$ is one half of the Dirichlet form of L_Δ plus a form bounded on L^2 ; the form domain $\{f : \int \log(2 + |t|)|F|^2 < \infty\}$ embeds compactly into $L^2(-a, a)$ [7, Thm. 2.1 ff.], so Q is bounded below and lower semicontinuous on the unit sphere and a minimizing sequence has a convergent subsequence. Density of C_c^∞ in the form domain gives the second claim. Monotonicity is inclusion of the admissible sets, and a monotone function is differentiable almost everywhere. Continuity: for $a < a'$ and a minimizer f' at a' , the dilate $f'_{a/a'}$ (notation of Section 5) is admissible at a , so $0 \leq \lambda(a) - \lambda(a') \leq Q(f'_{a/a'}) - Q(f')$. Minimizers at supports in $[a, a+1]$ have form norm bounded in terms of a and $\lambda(a)$ alone (the wall is positive, the prime and polar terms are bounded on the unit sphere), and on a form-bounded set $Q(f_b) - Q(f) \rightarrow 0$ uniformly as $b \rightarrow 1$: in the archimedean term because $|\Psi_\infty(s/b) - \Psi_\infty(s)| \leq w_* |\log b|$ with $w_* = \sup_t |t\Psi'_\infty(t)| < \infty$, in the prime terms because $|g_f(u) - g_f(u')| \leq \|f\| \|f(\cdot - u) - f(\cdot - u')\|$ and $\|f(\cdot - h) - f\|^2 \leq h^2 T^2 \|f\|^2 + 4 \int_{|t|>T} |F|^2$ is uniformly small on a form-bounded set, in the polar term because P is a bounded functional. Letting $a' \downarrow a$ gives right continuity; left continuity is $\lambda(a') \leq Q(f'_{a'/a}) \rightarrow Q(f) = \lambda(a)$ as $a' \uparrow a$, for a minimizer f at a . In particular the floor cannot jump. $\square$

2 Three engines

All computations use the orthonormal odd Legendre basis on the window, $f(y) = a^{-1/2} \sum_{i < K} c_i N_i P_{2i+1}(y/a)$, $N_i = \sqrt{(4i+3)/2}$, in which the matrix $Q_K(a)$ of (1) is assembled from the exact cross-correlations $g_{ij}(x) = \int \phi_i(u) \phi_j(u-x) du$ by Gauss–Legendre rules exact for the polynomial products. The K -mode floor $\lambda_K(a)$ is the lowest eigenvalue of $Q_K(a)$; it is an upper bound on $\lambda(a)$, and λ_1 below denotes the second eigenvalue. Three implementations are used, all in the repository accompanying this paper [19].

1. **rh_weil_odd.py**: mpmath, 50–60 digits, $K \leq 72$, 80 outer quadrature nodes. Reproduces the floors and the failure points of [16] to 4–6 digits. Its outer rule is exact only to polynomial degree 159, which costs 10^{-4} in relative accuracy at $K = 72$.
2. **rh_weil_arb.py**: ball arithmetic (python-flint 0.9.0 over FLINT/Arb), $K \leq 230$, 500–800 bits, with certified Gauss–Legendre rules, $2K + 8$ inner and $2K + 100$ outer nodes, and a rigorous enclosure of the lowest eigenvalue of the assembled matrix. The Legendre recurrence is evaluated with a midpoint reset at each degree, since its ball version doubles the radius per degree; the enclosure therefore covers the assembly and the eigenvalue computation but not the polynomial evaluations, which are 500-bit floating point, nor the truncation of the outer rule. It reproduces the mpmath floors to twenty digits where both are converged and runs a 180-mode point in six minutes.
3. **rh_edge_fem.py**: odd-extended hat functions on a mesh of the half window graded geometrically to $10^{-12}a$ (350–490 nodes; the “two-stage” mesh has ratio 0.9 from $0.05a$ to $10^{-4}a$ and $\frac{1}{2}$ below), exact cross-correlations, closed-form archimedean tails, double precision. Built to read the minimizer at the edge, where a Legendre truncation resolves only $\delta \gtrsim a/K^2$. Its amplitudes carry a mesh bias of 10^{-3} (prime-free) to 10^{-2} (full form).

Computation 2.1. $\lambda_{40}(0.8) = 1.5935 \times 10^{-14}$ and $\lambda_{64}(0.8) = 1.5659 \times 10^{-14}$, both inside Zhu's certified odd interval of Theorem 1.3, which fixes the normalization and shows the truncation error at that support. At $a = 0.3$, where no prime is inside the window, $\lambda_{64}(0.3) = 0.2225569$ and the edge-FEM floor 0.2225566 agree to 1.3×10^{-6} ; at $a_3 = \frac{1}{2} \log 3$ the 64-mode profile and the FEM profile agree to 10^{-4} for $|y| \leq 0.99a$.

3 The archimedean wall and the edge law

Proposition 3.1 (decomposition on the window). *Let $J_\Gamma(s) = e^{-|s|/2}/(1 - e^{-2|s|})$, $\mathcal{J}(x) = \int_x^\infty J_\Gamma = \operatorname{artanh}(e^{-x/2}) + \arctan(e^{-x/2})$, and $c_1 = -(\gamma + \log \pi) - \frac{\pi}{2} - 3 \log 2$. For f supported in $[-a, a]$,*

$$W_\infty(g_f) = c_1 \|f\|^2 + \frac{1}{2} \iint_{[-a, a]^2} J_\Gamma(u-v) |f(u) - f(v)|^2 du dv + \int_{-a}^a V_a(y) |f(y)|^2 dy, \quad V_a(y) = \mathcal{J}(a-y) + \mathcal{J}(a+y),$$

and $V_a(y) = \frac{1}{2} \log \frac{1}{a-|y|} + \kappa_a + o(1)$ as $|y| \rightarrow a$, with $\kappa_a = \frac{1}{2} \log 4 + \frac{\pi}{4} + \mathcal{J}(2a)$.

Proof. With $g(0) - g(x) = \frac{1}{2} \int |f(u) - f(u-x)|^2 du$, the integral in (2) is $\frac{1}{2} \iint_{\mathbb{R}^2} J_\Gamma(u-v) |f(u) - f(v)|^2$, and $2 \int_0^\infty (e^{-2x} - e^{-x/2})/(1 - e^{-2x}) dx = \psi(\frac{1}{4}) + \gamma = -\pi/2 - 3 \log 2$ by the integral representation of ψ . Splitting the double integral according to whether v lies in the window gives the three terms, the two cross-regions contributing $\frac{1}{2} \int V_a |f|^2$ each. The substitution $w = e^{-s/2}$ gives $\int_x^\infty J_\Gamma = \int_0^{e^{-x/2}} 2 dw/(1 - w^4) = \mathcal{J}(x)$, and $\operatorname{artanh}(e^{-x/2}) = \frac{1}{2} \log(4/x) + o(1)$ gives the constant; at $a = a_3$, $\kappa_a = 2.660623$, and the direct evaluation of $V_a - \frac{1}{2} \log(1/\delta)$ at $\delta = 10^{-9}$ gives 2.6606231. $\square$

The symbol of the whole-line form $\frac{1}{2} \iint J_\Gamma |f(u) - f(v)|^2$ is $\sigma(\xi) = \int J_\Gamma(s)(1 - \cos s\xi) ds = \log |\xi| + O(1)$, so on the window the archimedean form is one half of the Dirichlet form $\int \log(\xi^2) |\hat{f}|^2$ of the logarithmic Laplacian L_Δ of Chen–Weth [7] plus a bounded quadratic form; the prime terms are bounded translation operators and the polar term has rank one. Write $Q(f) = \frac{1}{2} \mathcal{E}_L(f) + \langle Bf, f \rangle$ with B bounded and self-adjoint on $L^2(-a, a)$ (its kernel is bounded on the window). The weak Euler–Lagrange equation of a minimizer is then

$$L_\Delta f = 2(\lambda - B)f \quad \text{in } (-a, a), \quad f = 0 \text{ outside}, \quad (4)$$

tested against $C_c^\infty(-a, a)$.

Hypothesis (H_∞). The minimizer f is bounded. *This is not known to follow from the form domain, which embeds in no L^p , $p > 2$; an L^∞ bound of De Giorgi–Moser type for L_Δ with a bounded lower-order perturbation would supply it. Numerically $\max |f| = 1.37$ at a_3 and 1.61 at $a = 0.3$.*

Theorem 3.2 (edge law; Chen–Weth [7], Thm. 1.11; Hernández-Santamaría–López Ríos–Saldaña [8], Thm. 1.1). *Assume (H_∞). Then f is continuous on $\mathbb{R}$ and there is C_0 with*

$$|f(y)| \leq C_0 \left(\log \frac{1}{a-|y|} \right)^{-1/2} \quad (|y| < a, \ a - |y| < 0.1).$$

In particular $f(\pm a) = 0$.

Proof sketch. By (4) and (H_∞), $f \in H(\Omega) \cap L^\infty(\mathbb{R})$ is a weak solution of $L_\Delta f = h$ with $h = 2(\lambda - B)f \in L^\infty(\Omega)$, $\Omega = (-a, a)$, which is a Lipschitz domain with the uniform exterior sphere condition. [7, Thm. 1.11] gives continuity and the rate $(\log(1/\delta))^{-\tau}$ for every $\tau < \frac{1}{2}$; [8, Thm. 1.1] gives $\tau = \frac{1}{2}$, with $\ell(r) = -1/\ln \min\{0.1, r\}$. $\square$

The exponent $\frac{1}{2}$ is sharp for the torsion function of a small interval [8, Thm. 1.2], and a Hopf-type lemma gives the rate from below for nonnegative supersolutions [8, Thm. 1.4]; the minimizer here is odd, so the lemma does not apply to it directly and the lower rate is a numerical statement (Computation 3.4).

Heuristic 3.3. In $t = \log(1/\delta)$, $\delta = a - y$, equation (4) near the edge has the shape $t h(t) - \frac{1}{2} \int_0^t h(\tau) d\tau = O(h) + O(1)$, the coefficient t being $\frac{1}{2}$ from the wall and $\frac{1}{2}$ from the interior kernel, and the nonlocal term coming from the long-range part of the kernel, which in the variable t is the constant $\frac{1}{2}$ on one side; its solutions are $h = Ct^{-1/2}(1 + O(1/t))$. We have not derived the equation in this form.

Checkpoint 17 of [17] applied Theorem 3.2 to the forced solutions of the same operator and left open the existence of the normalized trace $\lim \sqrt{\log(1/\delta)} f(a - \delta)$; the following computation is the numerical answer for the ground state.

Computation 3.4 (the edge amplitude). On every mesh and with or without primes, $1/f(a - \delta)^2$ is linear in $t = \log(a/\delta)$ with relative residual below 10^{-4} over $14 \leq t \leq 20$; the local exponent $-d \log f / d \log t$ descends toward $\frac{1}{2}$ exactly as $\frac{1}{2}t/(t + \beta)$ does. Writing $f(a - \delta) = C(t + \beta)^{-1/2}(1 + o(1))$ on the two-stage mesh:

form	a	λ (FEM)	C	β	mesh dependence of C
prime-free	0.3	0.2225566	1.3089	-1.49	10^{-3}
full	$a_3 = 0.5493$	1.4914×10^{-5}	0.02412	-1.87	10^{-2}

The six K -mode endpoint values of [16] at a_3 ($f_K(a_3) = 0.009471, \dots, 0.008207$ for $K = 24, \dots, 64$) are this law read at the truncation's resolution: with the fitted C, β , the effective $t_K = C^2/f_K^2 - \beta$ equals $2 \log K + 2.0$ within 0.3 for all six, with and without primes, so a degree- $2K$ expansion resolves $\delta \approx e^{-2}a/K^2$ at the edge and $f_K(a)^2 = C^2/(2 \log K + \beta')$ with $\beta' = 2 + \beta$.

4 The relay is a soft kink

At an entry a_n the new term $-2\Lambda(n)n^{-1/2}g(\log n)$ vanishes and grows with a .

Proposition 4.1 (K -mode kink). *For the K -mode form, λ_K is continuous at a_n and $\lambda'_K(a_n^+) - \lambda'_K(a_n^-) = 4\Lambda(n)n^{-1/2}f_K(a_n)^2$, where f_K is the normalized K -mode minimizer at entry.*

Proof. For $a = a_n + \varepsilon$ and odd f , $g(\log n) = \int_{a-2\varepsilon}^a f(y)f(y - \log n) dy = - \int_0^{2\varepsilon} f(a - \delta)f(a - 2\varepsilon + \delta) d\delta$, which is $-2\varepsilon f(a)^2 + O(\varepsilon^2)$ for a function continuous at the edge; the K -mode minimizer is a polynomial. Hellmann–Feynman with the minimizer frozen gives the jump. $\square$

Computation 4.2 (the K -mode kink). With 24 modes at the entry of 3, the measured jump is 0.948 of the prediction 2.276×10^{-4} at step 5×10^{-4} and 0.9998 at step 10^{-5} (`rh_kink_steps.py` in [19]); the deficit at coarser steps is the reorganization of the minimizer within the scale gap/ $\|Q'_K\| \approx 3 \times 10^{-5}$, the ratio of the spectral gap $\lambda_1 - \lambda_K$ to the norm of the entering term's derivative.

For the exact form, under (H_∞) , Theorem 3.2 gives $f(a_n) = 0$ and the proposition's jump is a quantity of the truncation. What replaces it follows from the fitted profile of Computation 3.4: with $f(a - \delta) = C(\log(a/\delta) + \beta)^{-1/2}$, the product integral is

$$-2\Lambda(n)n^{-1/2}g(\log n) = \frac{4\Lambda(n)C^2}{\sqrt{n}} \frac{\varepsilon}{\log(a/\varepsilon) + \beta + 1 - \log 2} \left(1 + O(\log^{-2}(1/\varepsilon))\right), \quad (5)$$

so λ' is continuous across the entry and its increment at scale ε is $4\Lambda(n)n^{-1/2}C^2/(\log(a/\varepsilon) + \beta + 1 - \log 2)$: a soft kink, vanishing only logarithmically. (An earlier version omitted the constant $1 - \log 2$ in the denominator, which is an 8% effect at $\varepsilon = 10^{-3}$.)

Computation 4.3 (the soft kink). On the edge FEM (134-node mesh, for which $C = 0.0241$, $\beta = -1.93$) at the entry of 3, with $D(\varepsilon) = [\lambda(a_3 + \varepsilon) - 2\lambda(a_3) + \lambda(a_3 - \varepsilon)]/\varepsilon$, all entries $\times 10^{-4}$:

ε	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
measured $D(\varepsilon)$	2.45	1.90	1.48	1.21	1.01
first-order energy of the entering term on the a_3 minimizer	3.42	2.13	1.59	1.27	1.06
law (5)	3.15	2.11	1.59	1.27	1.06

The law reproduces the first-order energy from $\varepsilon = 10^{-4}$ down, as it must, since it is that quantity evaluated on the fitted profile; the measured increment sits 3–5% below both, which is the second-order reorganization of the minimizer, and at 10^{-3} the smooth curvature $\lambda''\varepsilon \approx 0.9 \times 10^{-4}$ is still inside D . The increment falls by about 22% per decade, as the law does. The 24-mode form gives the constant 2.28×10^{-4} .

At the resolution of the grids of Section 6 (steps 0.004 to 0.025) the entries beyond $n = 3$ leave no visible trace in λ'_K . The Hellmann–Feynman derivative resolves what those steps average over.

Computation 4.4 (the relay resolved). `rh_dilation.py` gives $\lambda'_K(a) = \langle c, Q'_K(a)c \rangle$ at $a_n + \text{offset}$ for the first five entries ($K = 48, 48, 64, 80, 96$ for $n = 2, 3, 4, 5, 7$; ball arithmetic, central differences of the matrix at step 10^{-30}). Entries are Φ'/T^* ; data in `relay_scan.csv`.

n	a_n	-0.01	-0.003	-0.001	-0.0001	+0.0001	+0.001	+0.01
2	0.3466	2.42	2.96	3.16	3.27	2.80	2.59	2.30
3	0.5493	2.31	3.02	3.65	4.08	3.41	3.17	3.08
4	0.6931	3.61	3.32	3.70	4.08	3.73	3.63	3.51
5	0.8047	3.96	2.98	3.32	4.32	3.63	3.48	3.51
7	0.9730	3.18	3.66	3.06	4.12	3.59	3.54	3.84

Over the last 0.003 before each entry the ratio rises by 10%, 35%, 23%, 45%, 13% ($n = 2, 3, 4, 5, 7$), monotonically for $n = 2, 3$ and after a dip at -0.003 for $n = 4, 5, 7$; across the entry, between offsets ∓ 0.0001 , it drops by 14%, 16%, 9%, 16%, 13%, which at this scale is the K -mode kink of Proposition 4.1 and the soft kink (5) nearly coincide, since $1/K^2 \approx 10^{-4}$; then it relaxes. The peaks at the entries are 3.27, 4.08, 4.08, 4.32, 4.12 and the dips between them reach 2.3. The values are stable in K : at $a_3 - 0.001$, $\Phi'/T^* = 3.6404, 3.6485, 3.6505$ for $K = 32, 48, 64$; at $a_5 - 0.003$, 3.001 at $K = 64$ against 2.98 at $K = 80$. The rise is the form without the entering prime power heading toward its failure point a_{fail} , the deleted-form failure of [16] (0.5574 for $n = 3$): λ_K falls roughly linearly there ($\lambda'_K \approx -1.14 \times 10^{-3}$ at $a = 0.549$, reaching zero at 0.562 by linear extrapolation) while λ'_K barely moves, so $\Phi' = -\lambda'_K/\lambda_K$ grows like $1/(a_{\text{fail}} - a)$ until the entry rescues the floor. Each value near a_3 was checked against a finite difference of λ_K with step 2×10^{-4} to four digits. The coarse grids average this sawtooth into the plateau of Computation 6.1.

The relay is therefore the sawtooth described in [16], measured on the K -mode form: not a kick at the entry but a rise into it, cut off by the entering prime power, whose first-order action at the entry is the soft kink.

5 The sliding-window identity

For f supported in $[-a, a]$ and $b > 0$ let $f_b(x) = b^{-1/2} f(x/b)$, supported in $[-ab, ab]$ with $\|f_b\| = \|f\|$. Define, with $F = \hat{f}$ and $g = g_f$,

$$D_\infty(f) = \frac{1}{2\pi} \int |F(t)|^2 t \partial_t \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) dt, \quad D_P(f) = - \sum_{\log n < 2a} \frac{2\Lambda(n)}{\sqrt{n}} \log n g'(\log n),$$

$P(f) = \int f(x) \sinh(x/2) dx$, $M(f) = \int f(x) x \cosh(x/2) dx$, and

$$\mathcal{D}(f) = D_\infty(f) + D_P(f) + 2P(f)^2 + 2P(f)M(f). \quad (6)$$

Lemma 5.1 (dilation derivative of the form). *Let a lie strictly between two consecutive entries and let f be supported in $[-a, a]$ with $Q(f) < \infty$ and g_f differentiable at every $\log n < 2a$. Then $b \mapsto Q(f_b)$ is differentiable at $b = 1$ and $\frac{d}{db} Q(f_b)|_{b=1} = -\mathcal{D}(f)$.*

Proof. $\hat{f}_b(t) = \sqrt{b} F(bt)$ exactly, jumps included. In (1), $g_{f_b}(u) = g_f(u/b)$, so the prime terms are $-2\Lambda(n)n^{-1/2}g_f(\log n/b)$, differentiable at $b = 1$ with derivative $2\Lambda(n)n^{-1/2} \log n g'_f(\log n) =$ the n -th term of $-D_P(f)$; for b near 1 no new prime enters. The archimedean term is $\frac{1}{2\pi} \int b |F(bt)|^2 \Psi_\infty(t) dt = \frac{1}{2\pi} \int |F(s)|^2 \Psi_\infty(s/b) ds$ with $\Psi_\infty = \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$; since $t\Psi'_\infty(t)$ is bounded on $\mathbb{R}$, differentiation under the integral is justified by dominated convergence and gives $-\frac{1}{2\pi} \int |F|^2 t\Psi'_\infty(t) dt = -D_\infty(f)$ at $b = 1$. For the polar term, $P(f_b) = \sqrt{b} \int f(y) \sinh(by/2) dy$ gives $\frac{d}{db} P(f_b)|_{b=1} = \frac{1}{2}P(f) + \frac{1}{2}M(f)$, so $\frac{d}{db}(-2P(f_b)^2)|_{b=1} = -2P^2 - 2PM$. $\square$

Theorem 5.2 (sliding-window identity, K -mode form). *Let a lie strictly between two consecutive entries and let $\lambda_K(a)$ be simple with normalized eigenvector c , f the corresponding function. Then λ_K is differentiable at a and*

$$a \lambda'_K(a) = -\mathcal{D}(f). \quad (7)$$

Proof. The basis functions are a dilation family, $\phi_i^{(ab)} = (\phi_i^{(a)})_b$, so $Q_K(ab) = [Q((\phi_i^{(a)})_b, (\phi_j^{(a)})_b)]_{ij}$ is real-analytic in b near 1 (each entry is a finite sum of terms as in Lemma 5.1, with g_{ij} polynomial), Hellmann–Feynman for a simple eigenvalue gives $\frac{d}{db} \lambda_K(ab)|_{b=1} = \langle c, \frac{d}{db} Q_K(ab)|_{b=1} c \rangle = \frac{d}{db} Q(f_b)|_{b=1}$ by bilinearity, and Lemma 5.1 evaluates it. $\square$

Proposition 5.3 (exact form: two-sided Dini derivatives). *Let a lie strictly between two consecutive entries and let f be a minimizer at support a (Lemma 1.4) with g_f differentiable at every $\log n < 2a$. Then*

$$a D^+ \lambda(a) \leq -\mathcal{D}(f) \leq a D_- \lambda(a),$$

where D^+, D_- are the upper-right and lower-left Dini derivatives; hence $a \lambda'(a) = -\mathcal{D}(f)$ at every point of differentiability of λ , which by Lemma 1.4 is almost every a .

Proof. For every $b > 0$, f_b is admissible at support ab , so $\lambda(ab) \leq Q(f_b)$, with equality at $b = 1$. By Lemma 5.1, $Q(f_b) = \lambda(a) - (b-1)\mathcal{D}(f) + o(b-1)$. For $b > 1$, $(\lambda(ab) - \lambda(a))/(b-1) \leq -\mathcal{D}(f) + o(1)$; for $b < 1$ the division reverses the inequality. Since $\frac{d}{db} = a \frac{d}{da}$ at $b = 1$, the two bounds follow. $\square$

Remark 5.4. The hypothesis on g'_f is implied by $f' \in L^1$, which the edge law $C(\log(1/\delta))^{-1/2}$ suggests but which is not proved; it is the only condition beyond existence. The form-domain condition $\int \log(2 + |t|)|F|^2 < \infty$ does not give $\int |t||F|^2 < \infty$, so the prime part of $\mathcal{D}$ is to be read in x -space, as defined.

The hypothesis is natural in the class of functions of bounded variation, and in that class the dilation derivative is bounded, which makes the floor Lipschitz.

Lemma 5.5 (dilation of a function of bounded variation). *Let $f \in L^\infty(\mathbb{R})$ have bounded variation $\text{Var } f$ and support in $[-a, a]$. Then g_f is Lipschitz with constant $\|f\|_\infty \text{Var } f$; the map $b \mapsto Q(f_b)$ is locally Lipschitz on $(0, \infty)$, entries included, with $b \frac{d}{db} Q(f_b) = -\mathcal{D}(f_b)$ at every b at which g_f is differentiable at the points $(\log n)/b$; and for every b*

$$|\mathcal{D}(f_b)| \leq w_* \|f\|^2 + \frac{\|f\|_\infty \text{Var } f}{b} \sum_{\log n < 2ab} \frac{2\Lambda(n) \log n}{\sqrt{n}} + (2\sigma(ab)^2 + 2\sigma(ab)\mu(ab)) \|f\|^2,$$

where $w_* = \sup_t t \partial_t \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) = 2.0807$, attained at $t = 0.520$, $\sigma(a)^2 = \int_{-a}^a \sinh^2(x/2) dx$ and $\mu(a)^2 = \int_{-a}^a x^2 \cosh^2(x/2) dx$. If f is continuous, $g_f \in C^1$ and $b \mapsto Q(f_b)$ is C^1 .

Proof. For f of bounded variation, $\int |f(y) - f(y-h)| dy \leq |h| \text{Var } f$, so $|g_f(u) - g_f(u')| \leq \|f\|_\infty \int |f(x-u) - f(x-u')| dx \leq \|f\|_\infty \text{Var } f |u-u'|$. The prime terms of $Q(f_b)$ are $-2\Lambda(n)n^{-1/2}g_f(\log n/b)$, locally Lipschitz in b , with $b \frac{d}{db} g_f(\log n/b) = -(\log n/b) g'_f(\log n/b) = -\log n g'_{f_b}(\log n)$ whenever g_f is differentiable at $\log n/b$; a term with $\log n = 2ab$ enters with the value $g_f(2a) = 0$ and continuously, since g_f vanishes on $[2a, \infty)$. The archimedean and polar terms are C^1 in b for every b , with the derivatives computed in Lemma 5.1. The bound uses $|t \partial_t \text{Re } \psi| \leq w_*$, $|g'_{f_b}| \leq \|f_b\|_\infty \text{Var } f_b = b^{-1} \|f\|_\infty \text{Var } f$, and $|P(f_b)| \leq \sigma(ab) \|f\|$, $|M(f_b)| \leq \mu(ab) \|f\|$ by Cauchy-Schwarz. If f is continuous, $g'_f(u) = -\int f(z+u) d(Df)(z)$ with Df the finite derivative measure of f , which is continuous in u by dominated convergence. $\square$

Proposition 5.6 (the floor is Lipschitz under a variation bound). *Let $0 < a_0 < a_1$ and $M > 0$, and assume*

(H_{BV}) for every $a \in [a_0, a_1]$ and $\varepsilon > 0$ there is $f \in L^\infty$ of bounded variation, supported in $[-a, a]$, with $\|f\| = 1$, $Q(f) \leq \lambda(a) + \varepsilon$ and $\|f\|_\infty \text{Var } f \leq M$.

Then for $a_0 \leq a < a' \leq a_1$,

$$0 \leq \lambda(a) - \lambda(a') \leq L \log \frac{a'}{a} \leq \frac{L}{a_0} (a' - a), \quad L = w_* + \frac{a_1}{a_0} M \sum_{\log n < 2a_1} \frac{2\Lambda(n) \log n}{\sqrt{n}} + 2\sigma(a_1)^2 + 2\sigma(a_1)\mu(a_1).$$

In particular λ is Lipschitz on $[a_0, a_1]$, hence absolutely continuous there, with $\lambda' \geq -L/a_0$ almost everywhere.

Proof. The left inequality is Lemma 1.4. For the right one take f' as in (H_{BV}) at a' and put $b_0 = a/a'$. The dilate f'_{b_0} is supported in $[-a, a]$ with unit norm, so $\lambda(a) \leq Q(f'_{b_0})$. By Lemma 5.5, $b \mapsto Q(f'_b)$ is Lipschitz on $[b_0, 1]$ with $|b \frac{d}{db} Q(f'_b)| \leq L$ almost everywhere (the prime sum runs over $\log n < 2a'b \leq 2a_1$, and $b^{-1} \leq a_1/a_0$), so $Q(f'_{b_0}) \leq Q(f') + L \log(1/b_0) \leq \lambda(a') + \varepsilon + L \log(a'/a)$. Let $\varepsilon \rightarrow 0$. $\square$

Remark 5.7. (H_{BV}) concerns near-minimizers, not the minimizer, and it follows from a bound on the K -mode minimizers uniform in K : $\lambda(a) = \inf_K \lambda_K(a)$ by Lemma 1.4 (the odd Legendre truncations of an odd $C_c^\infty(-a, a)$ function converge to it in the form norm, their jumps at $\pm a$ being of vanishing size and costing $O(\text{jump}^2)$ in a norm that tolerates jumps), so the K -mode minimizers are near-minimizers of the exact problem, and they are polynomials, hence of bounded variation. Under (H_{BV}) on every compact subinterval of $[a_0, \infty)$ the regularity clause of Conjecture 7.1 holds

and the conjecture is the inequality $\lambda' \geq -cT^*\lambda$ a.e. alone. The trivial bound $\lambda' \geq -L/a$ also shows what the conjecture asks: on $[0.6, 1]$ with $M = 9$ (Computation 5.8) it gives $L = 126$ and $\lambda' \geq -211$, while λ' is -4×10^{-5} at $a = 0.6$ and -3×10^{-24} at $a = 1$; the conjecture is that $\mathcal{D}(f)$, whose three terms are each of order one, cancels to $O(T^*\lambda)$ on the minimizer, as in (8).

Computation 5.8 (bounded variation of the minimizers). `rh_bv_check.py`. On the edge-FEM minimizers of Computation 8.2 (486 nodes, edge resolved to $10^{-12}a$, $\|f\| = 1$): the supremum, the total variation over the window, and the increments of the Hardy integral $H(\delta_0) = \int_{\delta_0}^{a/10} f(a - \delta)^2 d\delta/\delta$ between $\delta_0 = 10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}$ (finiteness of $H(0)$ is the $H^{1/2}$ condition at the edge).

a	primes	$\sup f $	$\text{Var } f$	$\text{Var } f / (4 \sup f)$	$H(10^{-3})$	increments per three decades
0.3	no	1.70138	6.80553	1.0000	2.64	1.63, 0.93, 0.62
0.4	no	1.47286	5.89146	1.0000	2.13	1.26, 0.69, 0.46
0.5	no	1.31669	5.26678	1.0000	1.83	1.01, 0.58, 0.39
0.6	no	1.20136	4.80545	1.0000	1.66	0.86, 0.52, 0.35
0.4	yes	1.55514	6.22057	1.0000	0.56	0.23, 0.12, 0.08
0.45	yes	1.54288	6.17152	1.0000	0.17	0.049, 0.027, 0.018
0.5	yes	1.49876	5.99842	1.0006	0.037	0.0054, 0.0030, 0.0020
a_3	yes	1.47916	5.91666	1.0000	0.0060	0.0005, 0.0003, 0.0002

On every minimizer $\text{Var } f = 4 \sup |f|$ to four digits (1.0006 at $a = 0.5$ with primes, where the half-window profile has a small interior ripple): each half-window profile rises from 0 at the centre to its maximum and falls monotonically to the edge. Mesh refinement at $a = 0.5$ without primes (124, 237, 471, 486 nodes; $\delta_{\min} = 6 \times 10^{-9}$ and then 7×10^{-13} ; 100, 200, 400, 200 uniform intervals on the inner 90% of the half window; the last mesh with the two-stage grading) gives $\sup |f| = 1.316720, 1.316696, 1.316694, 1.316694$ and $\text{Var } f = 5.266880, 5.266785, 5.266777, 5.266777$. The Hardy increments decrease like those of $C^2 \log(\log(a/\delta_0) + \beta)$ with the C, β of Computation 8.2 (at $a = 0.3$: 1.63, 0.93, 0.62 against 1.66, 0.83, 0.55), i.e. $H(\delta_0) \rightarrow \infty$ doubly logarithmically as the edge law predicts: the minimizers are not in $H^{1/2}$, and $L^\infty \cap BV$ is the right class. On the K -mode minimizers (ball-arithmetic engine, two steps of inverse iteration, evaluated on 44,000 points of the half window graded to $10^{-12}a$; the variation includes the jump $f_K(a)$ at the edge):

a	K	12	24	48	72	96	128
0.6	$\sup f_K $	1.461834	1.461771	1.461786	1.461780	1.461771	
	$\text{Var } f_K$	5.847337	5.847085	5.847145	5.847121	5.847085	
0.8	$\sup f_K $	1.407373	1.421807	1.421859	1.421851 [†]	1.421844	1.421842
	$\text{Var } f_K$	5.629491	5.687230	5.687435	5.687404 [†]	5.687377	5.687368

([†]: $K = 64$.) At $a = 1.0$ (600 bits): $\sup |f_K| = 1.392228, 1.399706, 1.399722, 1.399716, 1.399714$ and $\text{Var } f_K = 5.568912, 5.598825, 5.598888, 5.598865, 5.598858$ for $K = 24, 48, 80, 120, 160$. In every case $\text{Var } f_K / (4 \sup |f_K|) = 1.0000$, and $\|f_K\|_\infty \text{Var } f_K$ converges, to 8.55, 8.09, 7.84 at $a = 0.6, 0.8, 1.0$: (H_{BV}) holds on these supports with $M = 9$, in the form of Remark 5.7.

Since $t \partial_t \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) \rightarrow 1$, $D_\infty(f) = \|f\|^2 + O(\int |F|^2 t^{-2})$; on a minimizer $\|f\| = 1$ while λ is 10^{-90} at $a = 1.5$, so (7) forces

$$D_P(f) = -D_\infty(f) - 2P^2 - 2PM + O(aT^*\lambda) : \quad (8)$$

the Euler product, differentiated and applied to the derivative of the minimizer's autocorrelation at the prime-power positions, cancels the archimedean dilation form to ninety digits. Together with the explicit formula on the same function, $\sum 2\Lambda(n)n^{-1/2}g(\log n) = W_\infty(g) - 2P^2 - \lambda$, both the values and the log-weighted derivatives of g_f at the prime powers are pinned by the archimedean side.

Computation 5.9 (the identity on the K -mode minimizer). `rh_dilation.py` differentiates the archimedean matrix A , the prime matrices Π_n and the polar matrix $ss^\top$ of $Q_K(a)$ by central differences with step 10^{-30} in ball arithmetic and contracts them with the eigenvector; every entry is a ball with the radius shown, and the last column is the sum, which is $a\lambda'_K$ by Theorem 5.2.

a	K	λ_K	$a\langle c, A'c \rangle$	$a\langle c, \Pi'c \rangle$	$-2a\langle c, (ss^\top)'c \rangle$	sum = $a\lambda'_K$	radius
0.6	48	5.9659×10^{-7}	-1.014313	+1.086974	-0.072687	-2.493408×10^{-5}	4×10^{-20}
0.8	64	1.5659×10^{-14}	-1.016605	+1.103966	-0.087361	$-1.295854 \times 10^{-12}$	4×10^{-27}
1.0	80	1.4820×10^{-26}	-1.018057	+1.114966	-0.096909	$-2.516852 \times 10^{-24}$	3×10^{-39}
1.25	140	1.5442×10^{-50}	-1.019178	+1.123585	-0.104407	$-1.233712 \times 10^{-48}$	2×10^{-63}

The three columns are $-D_\infty$, $-D_P$ and $-(2P^2 + 2PM)$ evaluated through the matrices; that their sum is $a\lambda'_K$ is Theorem 5.2, not an independent check. The independent check is on the Fourier side: at $K = 12$, $a = 0.6$ (`rh_fourier_check.py`), with the same coefficient vector, D_∞ evaluated as $(1/2\pi) \int |F|^2 t \partial_t \Re \psi(\frac{1}{4} + \frac{it}{2}) dt$ with F in closed form through spherical Bessel functions and the Parseval tail, D_P from $g'(\log 2)$ and $g'(\log 3)$ of the exact autocorrelation, and P, M by quadrature, gives $D_\infty = 1.01430470102641$, $D_P = -1.08691690953774$, $2P^2 + 2PM = 0.0726373685293583$ and $-\mathcal{D}(f) = -2.51600180293 \times 10^{-5}$, against $a\langle c, Q'_K c \rangle = -2.51600180295 \times 10^{-5}$ from the matrices: twelve digits. (The eigenvector enters through one step of inverse iteration from a shift at relative distance 2^{-20} below λ_K , which fixes $a\langle c, Q'_K c \rangle$ to about seven digits: at $K = 12$ a second step moves it by 1×10^{-7} relatively, to $-2.51600203711 \times 10^{-5}$, which is the central difference of λ_K itself to twelve digits. The ball radii in the table bound the arithmetic only.) The archimedean dilation form sits at 1.014–1.019, the excess over $\|f\|^2$ being the finite- t correction; the prime dilation form grows with the primes present and cancels it; the polar term grows from 7% to 10%. The implied Φ'/T^* are 3.339, 3.324, 3.658, 3.774; at $a = 0.8$ this lies inside the sawtooth of the entry of 5 (Computation 4.4).

5.1 The derivative is a property of the ground state

Let $Q'_K(a)$ be the a -derivative of the K -mode matrix, a symmetric form on the K -mode space. Is there an inequality $-Q'_K \leq cT^* Q_K$ between the two forms, which would give the derivative bound for every f at once? Since a form inequality on the exact form would restrict to every K -mode subspace, the pencil $(-Q'_K, Q_K)$ answers this for the exact form too.

Computation 5.10 (the pencil). Largest and smallest eigenvalues of $Q_K^{-1}(-Q'_K)$ divided by T^* , computed from the midpoint matrices with an approximate QR eigenvalue routine at 500 bits (Q_K has condition number up to 10^{26} here, leaving over a hundred digits):

a	minimizer direction	largest	second, third	smallest
0.6	3.34	74.7	4.26, 3.69	-70.5
0.8	3.32	1.46×10^5	332, 6.08	-1.46×10^5
1.0	3.66	6.2×10^{10}	3.1×10^7 , 8.2×10^4	-6.2×10^{10}

The extreme eigenvalues come in $\pm$ pairs and grow explosively. They are near-null directions of Q_K whose dilation derivative is enormous relative to their value; heuristically they are the directions along which the minimizer rotates as a moves, and their size is of the order of the inverse of the scale gap/ $\|Q'_K\|$ of Computation 4.2. The ground state is neither extreme; it is a special interior direction in which the dilation derivative is anomalously small, by ten orders at $a = 1$. Hence the derivative bound of Section 7 is a statement about the ground-state eigenvector alone, $\lambda' = \langle f, Q'f \rangle$, and cannot be an inequality between forms. A proof must pass through the Euler–Lagrange equation (4) of the minimizer.

6 The floor to height 130

Computation 6.1 (the band). Three data sets (`floor_grid_K40.csv`, `floor_grid_K72.csv`, `floor_grid_arb.csv`): $K = 40$, step 0.0125 on $[0.30, 1.20]$ with the entries $a_n \pm 0.004$ added; $K = 72$, step 0.025 on $[0.475, 1.05]$; and the ball-arithmetic engine at pairs $a \pm 0.005$ around the centres 1.06, 1.10, 1.15, 1.25, 1.33, 1.40, 1.45, 1.50 with $K = 110, 110, 110, 140, 140, 180, 180, 180$, plus $K = 180$ at 1.25 and $K = 230$ at 1.5. Φ' is a centered difference over the nearest admissible pair (± 0.025 for $K = 72$, ± 0.005 for arb), never across an entry; the pair at 1.10 straddles $a_9 = 1.0986$ and is excluded. Every value is a K -mode upper bound on the floor. Convergence: λ_{40} agrees with λ_{72} to 5% up to $a = 0.925$ and 26% at 1.0; λ_{140} with λ_{180} to 2% at 1.25; at 1.5, λ_{180} and λ_{230} differ by 39% and the ratio by 4%, more modes raising it, so the last ratio is a lower bound. For $a > 1$ each eigenvalue carries a rigorous enclosure of the assembled matrix’s lowest eigenvalue: radius 2×10^{-64} for the four points computed before the midpoint reset of Section 2, $\approx 10^{-143}$ at 500 bits after it, and $\approx 10^{-233}$ at 800 bits; the enclosures do not cover the polynomial evaluations or the outer rule.

a	K	T^*	λ_K	Φ'	Φ'/T^*
0.500	72	17.1	1.938×10^{-4}	55.2	3.23
0.650	72	23.1	1.837×10^{-8}	74.8	3.24
0.775	72	29.6	2.105×10^{-13}	107.5	3.63
0.900	72	38.0	7.176×10^{-20}	138.5	3.64
1.000	72	46.4	1.491×10^{-26}	172.6	3.72
1.060	110	52.4	2.97×10^{-31}	195.8	3.74
1.150	110	62.7	1.04×10^{-39}	240.3	3.83
1.250	180	76.5	3.59×10^{-51}	289.7	3.79
1.330	140	89.8	3.58×10^{-62}	339.2	3.78
1.400	180	103.3	1.73×10^{-73}	398.4	3.86
1.450	180	114.2	1.22×10^{-82}	443.5	3.88
1.500	230	126.2	2.97×10^{-93}	491.1	3.89

For the $K = 72$ rows λ_K is the grid value at the centre; for the arb rows it is the geometric mean of the pair. Means of Φ'/T^* : 3.23 on $[0.5, 0.7]$ (5 grid points), 3.61 on $[0.85, 1.05]$ (5), 3.79 on $[1.06, 1.25]$ (3), 3.85 on $[1.33, 1.5]$ (4). One-parameter least-squares fits of Φ'/T^* against a on the nine tabulated rows with $a \geq 0.85$, residual = rms of the relative deviations: $\Phi' = cT^*$ with $c = 3.79$, 2.1%; $\Phi' = cT^* \log T^*$, 7.5%; the raw count $\Phi \propto N(T^*)$, whose ratio is $\propto 4a - 1$, 18.4%; the derivative of Zhu’s even-sector law $\Phi \simeq C N(T^*)/\log N(T^*)$, whose ratio is $\propto 2a(L - 1)/L^2$ with $L = \log N(T^*) = 2a + \log(2a - 1)$ and tends to C/π , 8.1%. With Zhu’s own $C = 2\pi^2$ that derivative is 3.14 at $a = 1$ and 3.72 at $a = 1.5$, a 42% rise against the observed 7%. The ratio

λ_1/λ_K stays above 5.7 and grows overall to 10^6 at $a = 1$ (with local decreases of up to 18%), so the K -mode ground state is simple throughout.

At step 0.004 the entries beyond $n = 3$ show no net drop in Φ' , because the difference quotient across an entry averages the steep rise before it against the drop at it; Computation 4.4 resolves both. Over the fine scan, Φ'/T^* lies in $[2.3, 4.3]$, and the supremum over all measured points, the quantity Conjecture 7.1 bounds, is 4.32, at the entry of 5.

7 The conjectures

Conjecture 7.1 (bounded relay). *There are $a_0 > 0$ and $c > 0$ such that λ is locally absolutely continuous on $[a_0, \infty)$ and*

$$\lambda'(a) \geq -cT^*(a)\lambda(a) \quad \text{for almost every } a \geq a_0.$$

The statement is meaningful whatever the sign of λ ; absolute continuity is part of the conjecture: Lemma 1.4 gives continuity and differentiability a.e., and Proposition 5.6 gives Lipschitz continuity under (H_{BV}) , a bound on the sup norm times the total variation of near-minimizers, which Computation 5.8 supports on every minimizer computed ($\text{Var } f = 4 \sup |f| \approx 6$, converged in the mesh and in K). Where $\lambda > 0$ it reads $\Phi' \leq cT^*$.

Proposition 7.2. *Conjecture 7.1 with $a_0 \leq 0.8$ implies the Riemann Hypothesis.*

Proof. Put $u(a) = \lambda(a) \exp(\frac{c}{2}T^*(a))$, which is locally absolutely continuous with $u' = e^{cT^*/2}(\lambda' + cT^*\lambda) \geq 0$ a.e., using $\frac{d}{da}T^* = 2T^*$. Hence u is non-decreasing and $\lambda(a) \geq \lambda(a_0) \exp(-\frac{c}{2}(T^*(a) - T^*(a_0)))$ for $a \geq a_0$. With $a_0 = 0.8$, Theorem 1.3 gives $\lambda(0.8) \geq 8.206 \times 10^{-15} > 0$, so $\lambda > 0$ on $[0.8, \infty)$, and $\lambda > 0$ on $(0, 0.8]$ by the same theorem. Theorem 1.2 concludes. $\square$

Conjecture 7.3 (decay law). *For $a \geq a_0$, $\lambda(a) > 0$ and $\Phi'(a)/T^*(a)$ is bounded above and below by positive constants. In its sharp form, $\Phi'(a)/T^*(a) \rightarrow c_0$, i.e. $\lambda(a) = \exp(-(c_0/2 + o(1))2\pi e^{2a})$.*

Computation 6.1 supports the weak form from height 17 to 130 and, on the odd sector, favours a constant $c_0 \approx 3.8$ (residual 2.1%) over every law with a logarithmic factor; the derivative of Zhu's even-sector law, whose constant $C = 2\pi^2$ would give the asymptote 2π , predicts a rise six times larger than observed. Whether the ratio saturates or continues to creep up beyond height 130 is not decidable from these data. Conjecture 7.1 is the one that implies the Riemann Hypothesis; the constant is not its content.

By Theorem 5.2 and Proposition 5.3, Conjecture 7.1 is, at almost every a , the inequality

$$\mathcal{D}(f) \leq c a T^*(a) Q(f) \quad \text{on the minimizer } f, \tag{9}$$

and by Computation 5.10 it holds only there.

Remark 7.4 (what a proof cannot be). Positivity of Q_{24} on a window is destroyed by displacing $\log 2$ by 3×10^{-14} at $a = 0.8$ and by displacements of order $\exp(-cT^*(a))$ in general [16, 19], so no statement about the primes that is invariant under such displacements can imply positivity; and Checkpoint 20 of [17] shows that bounding the reflected prime terms separately from the polar term yields $-\infty$ on translated tests. A proof of Conjecture 7.1 must therefore use the exact multiplicative structure of Λ , and by Computation 5.10 it must use that f minimizes Q .

7.1 Under the Riemann Hypothesis

Let PW_a be the Paley–Wiener space of transforms $F = \hat{f}$ of $f \in L^2(-a, a)$, here with f odd. If all zeros are real, $Q(f) = \sum_{\gamma} |F(\gamma)|^2$ (the sum over γ with multiplicity, of both signs; it converges when $Q(f) < \infty$), and the dilation computation of Lemma 5.1 on the zero side, $Q(f_b) = \sum_{\gamma} b |F(b\gamma)|^2$, gives at a point of differentiability

$$a\lambda' = \lambda + \sum_{\gamma} \gamma \partial_t |F|^2(\gamma)$$

whenever the last sum converges absolutely, so Conjecture 7.1 reads $\sum_{\gamma} \gamma \partial_t |F|^2(\gamma) \geq -(1 + caT^*) \sum_{\gamma} |F(\gamma)|^2$: the minimizer is small at the zeros to the same order in its derivative as in its value. And

$$\lambda(a) = \inf \{ \sum_{\gamma} |F(\gamma)|^2 : F \in \text{PW}_a \text{ odd, } \|f\|_2 = 1 \}$$

is the lower sampling constant of the zeros for PW_a . The zeros have density $(2\pi)^{-1} \log(t/2\pi)$, which exceeds a/π exactly above $T^*(a)$, so only the zeros above the horizon can sample PW_a , and the floor measures how well a band-limited function can hide from them.

Both statements can be evaluated on the zeros themselves. The explicit formula is unconditional, and every zero used below lies on the line, so for the K -mode minimizer the sum over the zeros of $|F_K(\gamma)|^2$ must return $Q(f_K) = \lambda_K$ up to the contribution of the zeros above the last one used; this closes the loop on the object the form was built from.

Computation 7.5 (the floor and the identity on the zeros). `rh_zero_side.py`, with the first 6000 zeros ($\gamma \leq 6365.85$; `data/zeros_6000.txt`, from `mpmath.zetazero`, the first 400 to 36 digits and the rest to 21). For the K -mode minimizers of Computation 5.9, F_K is evaluated in closed form through spherical Bessel functions at 35 digits; the sums run over $\gamma > 0$ with the factor 2; the tail above γ_{6000} is the Riemann–von Mangoldt density integrated against $|F_K|^2$, with $|F_K(t)|^2$ replaced by its mean $2f_K(a)^2/t^2$ beyond $\gamma_{6000} + 60$, and the derivative tail is the corresponding $-2\times$ that; $a\lambda'_K$ is the matrix-side value of Computation 5.9, the sums run over $\gamma > 0$.

a	K	λ_K	$2 \sum F_K(\gamma) ^2 + \text{tail}$	ratio	$a\lambda'_K$	$\lambda_K + 2 \sum \gamma \partial_t F_K ^2 + \text{tail}$	ratio
0.6	48	5.96594×10^{-7}	5.96554×10^{-7}	0.99993	-2.49341×10^{-5}	-2.49310×10^{-5}	0.99988
0.8	64	1.56587×10^{-14}	1.56568×10^{-14}	0.99988	-1.29585×10^{-12}	-1.29685×10^{-12}	1.00077
1.0	80	1.48201×10^{-26}	1.48157×10^{-26}	0.99971	-2.51685×10^{-24}	-2.52433×10^{-24}	1.0030

At $a = 0.6$ the value ratio with $N = 10, 30, 100, 300, 1000, 3000, 6000$ zeros (tail included) is 0.977, 0.952, 0.973, 0.989, 0.9974, 0.99967, 0.99993; the residual is the fluctuation of the zeros about their mean density, of the size of the tail. The derivative sum is first-order sensitive to the positions of the zeros, by $4\gamma |F'_K(\gamma)|^2 \varepsilon$ per zero for an error ε in γ , while the value sum is not; with all zeros at 21 digits the derivative sum at $a = 1$ came out $+7 \times 10^{-22}$ against -2.5×10^{-24} , stable from $N = 30$ on, and the 36-digit zeros cure it. The residuals of the derivative sums, 1.2×10^{-4} , 8×10^{-4} , 3×10^{-3} , grow with a as this sensitivity does (the zeros beyond the 400th are still at 21 digits); the value sums are insensitive, and their residuals are the fluctuation of the zeros about their mean density in the tail. The explicit formula, which built the form from the primes and the archimedean symbol, returns on the actual zeros both the floor and its rate of change.

Where the floor sits among the zeros is the second reading of the same computation. At $a = 0.6$ ($T^* = 20.9$) one zero lies below the horizon and carries 1.0×10^{-5} of $2 \sum |F_K(\gamma)|^2$; at $a = 0.8$ ($T^* = 31.1$) four zeros lie below and carry 1.0×10^{-7} ; at $a = 1$ ($T^* = 46.4$) eight lie below and

carry 1.6×10^{-13} , the first zero alone 9×10^{-25} . The largest single terms are at $\gamma_7 = 40.9$ (15% of the floor), $\gamma_{13} = 59.3$ (19%) and $\gamma_{27} = 94.7$ (8%), at 2.0, 1.9 and 2.0 horizons, and at each support the mass sits between 1.5 and 3 horizons. So the minimizer hides from the zeros below the horizon to an accuracy far beyond the floor itself, and the floor is made by the zeros between the horizon and about three times the horizon, as the sampling picture says. This is the shadow in its most literal form: $2 \sum_{\gamma > 0} |F_K(\gamma)|^2 = \lambda_K$ says that the worst test function is small at every zero, to 10^{-13} at $a = 1$, and the derivative column says that it is small there in its dilation derivative too.

Heuristic 7.6 (prolate lower bound). The number of zeros in $[-T^*, T^*]$ is $2N(T^*) \approx (T^*/\pi)(2a - 1)$, short of the Shannon number $2aT^*/\pi$ of $[-a, a] \times [-T^*, T^*]$ by T^*/π . By the Landau–Pollak–Slepian theory [12, 13], with the asymptotics of the prolate spectrum in the plunge region [14, 15] (about $(2/\pi^2) \log c$ eigenvalues between ε and $1 - \varepsilon$ for $c = aT^*$), a function in PW_a constrained to vanish on all zeros below the horizon can still keep all but $\eta \gtrsim \exp(-\pi T^*/(\log(aT^*) + O(1)))$ of its energy below the horizon; under RH this bounds λ from above, i.e. $\Phi \gtrsim \pi T^*/\log(aT^*)$, which is $0.7 T^*$ at $a = 1$. The observed value is $\Phi(1) = 59.5 = 1.28 T^*$ (Computation 6.1); the zeros are about twice as expensive a constraint set as the count alone predicts, and the difference is arithmetic. Conjecture 7.1 is the matching upper bound.

8 The boundary law

The derivative of the floor should be a boundary quantity. Since Q does not depend on a , $\lambda(a)$ is the infimum of a fixed form over the growing family $L^2(-a, a)$, and its derivative is a domain variation. For the Dirichlet fractional Laplacian, Hadamard’s formula expresses such a derivative through the boundary trace u/dist^s of the eigenfunction [10]; for the logarithmic Laplacian the boundary trace is the amplitude C of Computation 3.4. A Pohozaev identity for L_Δ with the local boundary term $\int_{\partial\Omega} u^2 \ln(\delta^{-2}) (x \cdot \nu)$ was claimed in [9], which its authors withdrew on 26 April 2025 “due to a crucial error in Proof of Pohozaev identity”; to our knowledge no correct version exists. We therefore state the law as a conjecture and give the evidence.

Conjecture 8.1 (boundary law). *Assume (H_∞) . For a not an entry, with $C_\pm(a) = \lim_{\delta \rightarrow 0} \sqrt{\log(1/\delta)} |f(\pm(a - \delta))|$ the edge amplitudes of the minimizer (assumed to exist),*

$$\lambda'(a) = -(C_+(a)^2 + C_-(a)^2) = -2C(a)^2.$$

Equivalently, with Proposition 5.3, the Weil form on a window satisfies the Pohozaev-type identity $\mathcal{D}(f) = 2a C(a)^2$ on its minimizer.

Computation 8.2 (the boundary law). C from the edge FEM (two-stage mesh to $10^{-12}a$, fit of $1/f^2$ against $\log(a/\delta)$ on $14 \leq t \leq 20$, residual $\leq 5 \times 10^{-5}$, mesh bias 10^{-3} to 10^{-2} in C) and λ'_{48} from `rh_dilation.py` (Hellmann–Feynman, ball arithmetic, 48 modes; the truncation error of λ'_{48} is below 10^{-4} at these supports by the K -stability of Computation 4.4); data in `boundary_law.csv` and `data/`, reproduced by `rh_hadamard.py`:

a	primes	λ (FEM)	C	β	λ'_{48}	$r = -\lambda'/2C^2$
0.3	no	+0.2225566	1.30891	-1.49	-3.42881	1.0007
0.4	no	-0.0780502	1.15425	-1.48	-2.66640	1.0007
0.5	no	-0.3222425	1.06222	-1.45	-2.25815	1.0007
0.6	no	-0.5358860	1.00932	-1.42	-2.03880	1.0007
0.4	yes	1.470510×10^{-2}	0.48266	-1.57	-0.465947	1.0001
0.45	yes	2.404578×10^{-3}	0.22494	-1.67	-0.101283	1.0009
0.5	yes	1.936801×10^{-4}	0.07493	-1.78	-1.123556×10^{-2}	1.0005
0.549	yes	1.491447×10^{-5}	0.02412	-1.87	-1.143529×10^{-3}	0.983

Seven cases give $r = 1$ to better than 10^{-3} , across supports where the prime-free floor is positive and negative and with the primes present; the prime-free offset 7×10^{-4} is the mesh bias of C on that family, and the full-form offsets $(1, 9, 5 \times 10^{-4})$ are within the stated bias. The last case measures C at $a_3 = 0.54931$ and λ' at 0.549, inside the sawtooth where Φ' changes by about 9% per 0.001 (Computation 4.4); its 1.7% is within the combined error of the amplitude (10^{-2} in C , hence 2% in C^2) and the offset in a . The evidence is confined to $a \leq 0.6$ (heights below 22), far from the regime of Conjecture 7.1; extending it needs a multiprecision edge engine.

An exactly solvable case: for one half of the Dirichlet form of L_Δ on $(-a, a)$, the scaling identity $L_\Delta(u(\cdot/a)) = (L_\Delta u)(\cdot/a) - 2 \log a u(\cdot/a)$ gives $\lambda(a) = \lambda(1) - \log a$, so the law predicts $C(a)^2 = 1/(2a)$. The prime-free form differs from this by the bounded corrections (the polar term and $\operatorname{Re} \psi - \log |t|$); its amplitudes give $2aC^2 = 1.028, 1.066, 1.129, 1.223$ at $a = 0.3, \dots, 0.6$ and its derivatives $-a\lambda' = 1.029, 1.067, 1.129, 1.223$, both near 1 and equal to each other, which is the content of the table restated, not an independent test.

Remark 8.3 (consequence of Conjecture 8.1). $\Phi'(a) = 2C(a)^2/\lambda(a)$, and Conjecture 7.1 becomes

$$C(a)^2 \leq \frac{\varepsilon}{2} T^*(a) \lambda(a) \quad (a \geq a_0) :$$

the edge amplitude of the minimizer is bounded by its energy times the horizon. This is the form in which the conjecture is a statement about one function at one place. The relay of Computation 4.4 reads: as a approaches an entry from below, λ falls toward the deleted form's failure point while C stays of the same order, and $\Phi' = 2C^2/\lambda$ rises until the entry rescues the floor. Under the Riemann Hypothesis, $C(a)$ is the boundary trace of the band-limited function that best hides from the zeros above the horizon.

9 Relation to the literature, and acknowledgement

Suzuki [4] organizes the Weil form on $[-a, a]$ as a family of self-adjoint operators and conjectures that an operator with the zeros as spectrum is their limit as $a \rightarrow \infty$; Conjecture 7.1 is a differential inequality for the ground state of that family. Zhu [5] certifies both parity sectors at $a = 0.8$; his Landau–Widom decay law is fitted to the even-sector floor, calibrated on $L \leq 1.2$ and validated at $L = 1.4, \dots, 2.0$, and his exploratory 950-mode value at $L = 1.19$, between 10^{-48} and 10^{-46} , is for the full space of test functions, i.e. the even floor; the odd floors of Section 6 interpolate to $\lambda(1.19) \approx 3 \times 10^{-44}$, and the even floor is already three orders below the odd one at $a = 0.8$ (Theorem 1.3), so the two are consistent. Comparisons between his law and the odd Φ' are cross-sector and are labelled so. Connes–Consani [6] treat the range $a \leq \frac{1}{2} \log 2$ where only the archimedean place acts; the wall of Proposition 3.1 is that place acting alone at the edge for every a . The atlas [17] contains an exact-rational certificate of the odd floor to $a = 3/8$ with constant $1/200$, inside Zhu's range,

and the first application of Theorem 3.2 to this operator. The folds paper [16] computed the floors to $a = 1.1$, the deleted-form failure points and the symbol’s negative set, and described the relay as a sawtooth, which Computation 4.4 measures; the Möbius companion [18] treats the Green-energy side of the same program.

Computation 7.5 is the converse use of the explicit formula: the form was built from the primes and the archimedean symbol, and evaluating the minimizer on the actual zeros returns the floor and its derivative; with Computations 5.9 and the Fourier-side check the identity has now been measured from the matrices, from the symbol and from the zeros. An independent referee report on the first version of this paper identified the gaps corrected here: the identity had been asserted for the exact form on numerical grounds (now Theorem 5.2 for the K -mode form and Proposition 5.3 for the exact one), Conjecture 7.1 had been phrased in terms of Φ and so presupposed $\lambda > 0$, the abstract overstated several computations, the resolution threshold had been misattributed, the scaling identity carried a sign error, (5) lacked the constant $1 - \log 2$, and Computation 6.1 misdescribed its data; the report’s own recomputations of every table from the data files are gratefully acknowledged.

Reproducibility

All scripts and data are in [19]: `rh_weil_odd.py`, `rh_weil_arb.py` (python-flint 0.9.0), `rh_edge_fem.py` (numpy 2), `rh_soft_kink.py`, `rh_floor_grid.py`, `rh_arb_grid.py`, `rh_dilation.py` (python3 `rh_dilation.py` a K [bits] [noprimess]), `rh_hadamard.py`, `rh_fourier_check.py` (the Fourier-side check of Computation 5.9), `rh_bv_check.py`, `rh_zero_side.py` (Computation 7.5, with the zeros in `data/zeros_6000.txt`), and the data `floor_grid_K40.csv`, `floor_grid_K72.csv`, `floor_grid_arb.csv`, `boundary_law.csv`, `relay_scan.csv` and the folder `data/` (edge-FEM outputs and dilation logs behind Computation 8.2).

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